

Substitution lemma with de Bruijn indices

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Definition 1.

Environments $(\Gamma, \Delta, \dots)$

$$\begin{aligned} \text{Env} &: \mathcal{U} \\ \emptyset &: \text{Env} \\ \bullet &: \text{Env} \rightarrow \text{Env} \end{aligned}$$

Variables $(x, y, \dots)$

$$\begin{aligned} \text{Var}(-) &: \text{Env} \rightarrow \mathcal{U} \\ \mathbf{Z} &: \text{Var}(\Gamma) \\ \mathbf{S} &: \text{Var}(\Gamma) \rightarrow \text{Var}(\bullet\Gamma) \end{aligned}$$

Terms $(t, s, \dots)$

$$\begin{aligned} \text{Tm}(-) &: \text{Env} \rightarrow \mathcal{U} \\ \cdot &: \text{Var}(\Gamma) \rightarrow \text{Tm}(\Gamma) \\ \lambda &: \text{Tm}(\bullet\Gamma) \rightarrow \text{Tm}(\Gamma) \\ @ &: \text{Tm}(\Gamma) \rightarrow \text{Tm}(\Gamma) \rightarrow \text{Tm}(\Gamma) \end{aligned}$$

Renamings $(\rho, \rho', \dots)$

$$\begin{aligned} \text{Ren}(-, -) &: \text{Env} \rightarrow \text{Env} \rightarrow \mathcal{U} \\ \text{Ren}(\Gamma, \Delta) &\stackrel{\text{def}}{=} \text{Var}(\Gamma) \rightarrow \text{Var}(\Delta) \\ \uparrow - &: \text{Ren}(\Gamma, \Delta) \rightarrow \text{Ren}(\bullet\Gamma, \bullet\Delta) \\ \uparrow \rho(\mathbf{Z}) &\stackrel{\text{def}}{=} \mathbf{Z} \\ \uparrow \rho(\mathbf{S}(x)) &\stackrel{\text{def}}{=} \mathbf{S}(\rho(x)) \\ \mathbf{r}(-, -) &: \text{Ren}(\Gamma, \Delta) \rightarrow \text{Tm}(\Gamma) \rightarrow \text{Tm}(\Delta) \\ \mathbf{r}(\rho, \underline{x}) &\stackrel{\text{def}}{=} \underline{\rho(x)} \\ \mathbf{r}(\rho, \lambda t) &\stackrel{\text{def}}{=} \lambda \mathbf{r}(\uparrow \rho, t) \\ \mathbf{r}(\rho, @(t, s)) &\stackrel{\text{def}}{=} @(\mathbf{r}(\rho, t), \mathbf{r}(\rho, s)) \end{aligned}$$

Substitutions $(\sigma, \sigma', \dots)$

$$\begin{aligned} \text{Sub}(-, -) &: \text{Env} \rightarrow \text{Env} \rightarrow \mathcal{U} \\ \text{Sub}(\Gamma, \Delta) &\stackrel{\text{def}}{=} \text{Var}(\Gamma) \rightarrow \text{Tm}(\Delta) \\ \uparrow\uparrow - &: \text{Sub}(\Gamma, \Delta) \rightarrow \text{Sub}(\bullet\Gamma, \bullet\Delta) \\ \uparrow\uparrow \sigma(\mathbf{Z}) &\stackrel{\text{def}}{=} \mathbf{Z} \\ \uparrow\uparrow \sigma(\mathbf{S}(x)) &\stackrel{\text{def}}{=} \mathbf{r}(\mathbf{S}, \sigma(x)) \\ \text{sub}(-, -) &: \text{Sub}(\Gamma, \Delta) \rightarrow \text{Tm}(\Gamma) \rightarrow \text{Tm}(\Delta) \\ \text{sub}(\sigma, \underline{x}) &\stackrel{\text{def}}{=} \sigma(x) \\ \text{sub}(\sigma, \lambda t) &\stackrel{\text{def}}{=} \lambda \text{sub}(\uparrow\uparrow \sigma, t) \\ \text{sub}(\sigma, @(t, s)) &\stackrel{\text{def}}{=} @(\text{sub}(\sigma, t), \text{sub}(\sigma, s)) \end{aligned}$$

If $f : A \rightarrow A$, we write $f^n : A \rightarrow A$ for the function given by $f^0(a) = a$ and $f^{n+1}(a) = f(f^n(a))$. By abuse of notation we usually write directly ρ instead of $\mathbf{r}(\rho, \cdot)$ and σ instead of $\text{sub}(\sigma, \cdot)$. We write $\mathbf{r}^n(\rho, t)$ for the result of iterating n times $\mathbf{r}(\rho, -)$, that is:

$$\mathbf{r}^0(\rho, t) \stackrel{\text{def}}{=} t \quad \mathbf{r}^{n+1}(\rho, t) \stackrel{\text{def}}{=} \mathbf{r}(\rho, \mathbf{r}^n(\rho, t))$$

Lemma 2 (Commutation of renaming for variables). $(\uparrow^{n+1} \rho)((\uparrow^n \mathbf{S})(x)) = (\uparrow^n \mathbf{S})((\uparrow^n \rho)(x))$

Proof. By induction on n :

1. $n = 0$: Then $(\uparrow \rho)(\mathbf{S}(x)) = \mathbf{S}(\rho(x))$.

2. $n = m + 1$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned}
 (\uparrow^{m+2} \rho)((\uparrow^{m+1} \mathbf{S})(\mathbf{Z})) &= (\uparrow^{m+2} \rho)(\uparrow(\uparrow^m \mathbf{S})(\mathbf{Z})) \\
 &= (\uparrow^{m+2} \rho)(\mathbf{Z}) \\
 &= \uparrow(\uparrow^{m+1} \rho)(\mathbf{Z}) \\
 &= \mathbf{Z} \\
 &= \uparrow(\uparrow^m \mathbf{S})(\mathbf{Z}) \\
 &= (\uparrow^{m+1} \mathbf{S})(\mathbf{Z}) \\
 &= (\uparrow^{m+1} \mathbf{S})(\uparrow^{m+1} \rho)(\mathbf{Z})
 \end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned}
 (\uparrow^{m+2} \rho)((\uparrow^{m+1} \mathbf{S})(\mathbf{S}(y))) &= (\uparrow^{m+2} \rho)(\uparrow(\uparrow^m \mathbf{S})(\mathbf{S}(y))) \\
 &= (\uparrow^{m+2} \rho)(\mathbf{S}(\uparrow^m \mathbf{S})(y))) \\
 &= \uparrow(\uparrow^{m+1} \rho)(\mathbf{S}(\uparrow^m \mathbf{S})(y))) \\
 &= \mathbf{S}(\uparrow^{m+1} \rho)(\uparrow^m \mathbf{S})(y))) \\
 &= \mathbf{S}(\uparrow^m \mathbf{S})(\uparrow^m \rho)(y))) \quad \text{by IH} \\
 &= \uparrow(\uparrow^m \mathbf{S})(\mathbf{S}(\uparrow^m \rho)(y))) \\
 &= (\uparrow^{m+1} \mathbf{S})(\mathbf{S}(\uparrow^m \rho)(y))) \\
 &= (\uparrow^{m+1} \mathbf{S})(\uparrow(\uparrow^m \rho)(\mathbf{S}(y))) \\
 &= (\uparrow^{m+1} \mathbf{S})(\uparrow^{m+1} \rho)(\mathbf{S}(y)))
 \end{aligned}$$

□

Lemma 3 (Commutation of renaming). $\mathbf{r}(\uparrow^{n+1} \rho, \mathbf{r}(\uparrow^n \mathbf{S}, t)) = \mathbf{r}(\uparrow^n \mathbf{S}, \mathbf{r}(\uparrow^n \rho, t))$.

In particular, when $n = 0$, we have $\mathbf{r}(\uparrow \rho, \mathbf{r}(\mathbf{S}, t)) = \mathbf{r}(\mathbf{S}, \mathbf{r}(\rho, t))$.

Proof. By induction on t :

1. $t = \underline{x}$:

$$\begin{aligned}
 \mathbf{r}(\uparrow^{n+1} \rho, \mathbf{r}(\uparrow^n \mathbf{S}, \underline{x})) &= \mathbf{r}(\uparrow^{n+1} \rho, \uparrow^n \mathbf{S}(\underline{x})) \\
 &= \frac{\uparrow^{n+1} \rho(\uparrow^n \mathbf{S}(\underline{x}))}{\uparrow^n \mathbf{S}(\uparrow^n \rho(\underline{x}))} \quad \text{by lemma 2} \\
 &= \frac{\mathbf{r}(\uparrow^n \mathbf{S}, \uparrow^n \rho(\underline{x}))}{\mathbf{r}(\uparrow^n \mathbf{S}, \uparrow^n \rho(\underline{x}))} \\
 &= \mathbf{r}(\uparrow^n \mathbf{S}, \mathbf{r}(\uparrow^n \rho, \underline{x}))
 \end{aligned}$$

2. $t = \lambda s$:

$$\begin{aligned}
 \mathbf{r}(\uparrow^{n+1} \rho, \mathbf{r}(\uparrow^n \mathbf{S}, \lambda s)) &= \mathbf{r}(\uparrow^{n+1} \rho, \lambda \mathbf{r}(\uparrow^{n+1} \mathbf{S}, s)) \\
 &= \lambda \mathbf{r}(\uparrow^{n+2} \rho, \mathbf{r}(\uparrow^{n+1} \mathbf{S}, s)) \\
 &= \lambda \mathbf{r}(\uparrow^{n+1} \mathbf{S}, \mathbf{r}(\uparrow^{n+1} \rho, s)) \quad \text{by IH} \\
 &= \mathbf{r}(\uparrow^n \mathbf{S}, \lambda \mathbf{r}(\uparrow^{n+1} \rho, s)) \\
 &= \mathbf{r}(\uparrow^n \mathbf{S}, \mathbf{r}(\uparrow^n \rho, \lambda s))
 \end{aligned}$$

3. $t = @ (s, u)$: Straightforward by IH.

□

Lemma 4. $\mathbf{r}(\uparrow^n \mathbf{S}, \mathbf{r}^n(\mathbf{S}, t)) = \mathbf{r}^{n+1}(\mathbf{S}, t)$

Proof. By induction on n :

1. $n = 0$: $\mathbf{r}(\uparrow^0 \mathbf{S}, \mathbf{r}^0(\mathbf{S}, t)) = \mathbf{r}(\uparrow^0 \mathbf{S}, t) = \mathbf{r}(\mathbf{S}, t) = \mathbf{r}^1(\mathbf{S}, t)$.

2. $n \implies n + 1$:

$$\begin{aligned}
 \mathbf{r}(\uparrow^{n+1} \mathbf{S}, \mathbf{r}^{n+1}(\mathbf{S}, t)) &= \mathbf{r}(\uparrow(\uparrow^n \mathbf{S}), \mathbf{r}(\mathbf{S}, \mathbf{r}^n(\mathbf{S}, t))) \\
 &= \mathbf{r}(\mathbf{S}, \mathbf{r}(\uparrow^n \mathbf{S}, \mathbf{r}^n(\mathbf{S}, t))) \quad \text{by lemma 3} \\
 &= \mathbf{r}(\mathbf{S}, \mathbf{r}^{n+1}(\mathbf{S}, t)) \quad \text{by IH} \\
 &= \mathbf{r}^{n+2}(\mathbf{S}, t)
 \end{aligned}$$

□

Definition 5 (Substitution of the first variable).

$$\begin{aligned}
 [\cdot] &: \text{Tm}(\Gamma) \rightarrow \text{Sub}(\bullet \Gamma, \Gamma) \\
 [t](\mathbf{Z}) &\stackrel{\text{def}}{=} t \\
 [t](\mathbf{S}(x)) &\stackrel{\text{def}}{=} \underline{x}
 \end{aligned}$$

Lemma 6 (Commutation of substitution/renaming for variables). $(\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) = \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(x))$.

Proof. By induction on n :

1. $n = 0$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) &= (\uparrow\sigma)(\mathbf{S}(\mathbf{Z})) \\ &= \mathbf{r}(\mathbf{S}, \sigma(\mathbf{Z})) \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(x)) \end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) &= (\uparrow\sigma)(\mathbf{S}(\mathbf{S}(y))) \\ &= \mathbf{r}(\mathbf{S}, \sigma(\mathbf{S}(y))) \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n[s])(x)) \end{aligned}$$

2. $n = m + 1$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) &= (\uparrow^{m+2}\sigma)((\uparrow^{m+1}\mathbf{S})(\mathbf{Z})) \\ &= (\uparrow^{m+2}\sigma)((\uparrow(\uparrow^m\mathbf{S}))(\mathbf{Z})) \\ &= (\uparrow^{m+2}\sigma)(\mathbf{Z}) \\ &= (\uparrow(\uparrow^{m+1}\sigma))(\mathbf{Z}) \\ &= \mathbf{Z} \\ &= (\uparrow(\uparrow^m\mathbf{S}))(\mathbf{Z}) \\ &= \frac{(\uparrow^{m+1}\mathbf{S})(\mathbf{Z})}{\mathbf{r}(\uparrow^{m+1}\mathbf{S}, \mathbf{Z})} \\ &= \mathbf{r}(\uparrow^{m+1}\mathbf{S}, (\uparrow(\uparrow^m\sigma))(\mathbf{Z})) \\ &= \mathbf{r}(\uparrow^{m+1}\mathbf{S}, (\uparrow^{m+1}\sigma)(\mathbf{Z})) \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(x)) \end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) &= (\uparrow^{m+2}\sigma)((\uparrow^{m+1}\mathbf{S})(\mathbf{S}(y))) \\ &= (\uparrow^{m+2}\sigma)((\uparrow(\uparrow^m\mathbf{S}))(\mathbf{S}(y))) \\ &= (\uparrow^{m+2}\sigma)(\mathbf{S}((\uparrow^m\mathbf{S})(y))) \\ &= (\uparrow(\uparrow^{m+1}\sigma))(\mathbf{S}((\uparrow^m\mathbf{S})(y))) \\ &= \mathbf{r}(\mathbf{S}, (\uparrow^{m+1}\sigma)((\uparrow^m\mathbf{S})(y))) \\ &= \mathbf{r}(\mathbf{S}, \mathbf{r}(\uparrow^m\mathbf{S}, (\uparrow^m\sigma)(y))) && \text{by IH} \\ &= \mathbf{r}(\uparrow(\uparrow^m\mathbf{S}), \mathbf{r}(\mathbf{S}, (\uparrow^m\sigma)(y))) && \text{by lemma 3} \\ &= \mathbf{r}(\uparrow^{m+1}\mathbf{S}, \mathbf{r}(\mathbf{S}, (\uparrow^m\sigma)(y))) \\ &= \mathbf{r}(\uparrow^{m+1}\mathbf{S}, (\uparrow(\uparrow^m\sigma))(\mathbf{S}(y))) \\ &= \mathbf{r}(\uparrow^{m+1}\mathbf{S}, (\uparrow^{m+1}\sigma)(\mathbf{S}(y))) \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(x)) \end{aligned}$$

□

Lemma 7 (Commutation of substitution/renaming). $(\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(t)) = \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(t))$.

In particular, when $n = 0$, we have $(\uparrow\sigma)(\mathbf{S}(t)) = \mathbf{r}(\mathbf{S}, \sigma(t))$.

Proof. By induction on t :

1. $t = \underline{x}$: Then:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(\underline{x})) &= (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(x)) \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(x)) && \text{by lemma 6} \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(\underline{x})) \end{aligned}$$

2. $t = \lambda u$:

$$\begin{aligned} (\uparrow^{n+1}\sigma)((\uparrow^n\mathbf{S})(\lambda u)) &= \lambda(\uparrow^{n+2}\sigma)((\uparrow^{n+1}\mathbf{S})(u)) \\ &= \lambda\mathbf{r}(\uparrow^{n+1}\mathbf{S}, (\uparrow^{n+1}\sigma)(\lambda u)) && \text{by IH} \\ &= \mathbf{r}(\uparrow^n\mathbf{S}, (\uparrow^n\sigma)(\lambda u)) \end{aligned}$$

3. $t = @(\underline{u}, r)$: Straightforward by IH.

□

Lemma 8 (Weakening for variables). $(\uparrow^n[s])(\uparrow^n\mathbf{S})(x) = \underline{x}$

Proof. By induction on n :

1. $n = 0$: Then $[s](\mathbf{S}(x)) = \underline{x}$.

2. $n = m + 1$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned} (\uparrow^{m+1}[s])(\uparrow^{m+1}\mathbf{S})(\mathbf{Z}) &= (\uparrow^{m+1}[s])(\uparrow(\uparrow^m\mathbf{S})(\mathbf{Z})) \\ &= (\uparrow^{m+1}[s])(\mathbf{Z}) \\ &= \uparrow(\uparrow^m[s])(\mathbf{Z}) \\ &= \underline{\mathbf{Z}} \end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned} (\uparrow^{m+1}[s])(\uparrow^{m+1}\mathbf{S})(\mathbf{S}(y)) &= (\uparrow^{m+1}[s])(\uparrow(\uparrow^m\mathbf{S})(\mathbf{S}(y))) \\ &= (\uparrow^{m+1}[s])(\mathbf{S}(\uparrow^m\mathbf{S}(y))) \\ &= \uparrow(\uparrow^m[s])(\mathbf{S}(\uparrow^m\mathbf{S}(y))) \\ &= \mathbf{r}(\mathbf{S}, \uparrow^m[s])(\uparrow^m\mathbf{S}(y)) \\ &= \mathbf{r}(\mathbf{S}, \underline{y}) \quad \text{by IH} \\ &= \underline{\mathbf{S}(y)} \end{aligned}$$

□

Lemma 9 (Weakening). $(\uparrow^n[s])(\mathbf{r}(\uparrow^n\mathbf{S}, t)) = t$

In particular, when $n = 0$, we have $[s](\mathbf{r}(\mathbf{S}, t)) = t$.

Proof. By induction on t :

1. $t = \underline{x}$: Then

$$\begin{aligned} (\uparrow^n[s])(\mathbf{r}(\uparrow^n\mathbf{S}, \underline{x})) &= (\uparrow^n[s])(\uparrow^n\mathbf{S}(\underline{x})) \\ &= (\uparrow^n[s])(\uparrow^n\mathbf{S}(\underline{x})) \\ &= \underline{x} \quad \text{by lemma 8} \end{aligned}$$

2. $t = \lambda u$:

$$\begin{aligned} (\uparrow^n[s])(\mathbf{r}(\uparrow^n\mathbf{S}, \lambda u)) &= (\uparrow^n[s])(\lambda \mathbf{r}(\uparrow^{n+1}\mathbf{S}, u)) \\ &= \lambda (\uparrow^{n+1}[s])(\mathbf{r}(\uparrow^{n+1}\mathbf{S}, u)) \\ &= \lambda u \quad \text{by IH} \end{aligned}$$

3. $t = t_1 t_2$: Straightforward by IH.

□

Lemma 10 (Substitution lemma for variables). $(\uparrow^n[u])(\uparrow^n[s])(x) = (\uparrow^n[[u](s)])(\uparrow^{n+1}[u])(x)$

Proof. By induction on n :

1. $n = 0$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned} (\uparrow^0[u])(\uparrow^0[s])(\mathbf{Z}) &= [u]([s](\mathbf{Z})) \\ &= [u](s) \\ &= [[u](s)](\mathbf{Z}) \\ &= [[u](s)](\underline{\mathbf{Z}}) \\ &= [[u](s)](\uparrow^0[u])(\mathbf{Z}) \\ &= (\uparrow^0[[u](s)])(\uparrow^{0+1}[u])(\mathbf{Z}) \end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned} (\uparrow^0[u])(\uparrow^0[s])(\mathbf{S}(y)) &= [u]([s](\mathbf{S}(y))) \\ &= [u](y) \\ &= [u](y) \\ &= [[u](s)](\mathbf{r}(\mathbf{S}, [u](y))) \quad \text{by lemma 9} \\ &= [[u](s)](\uparrow^0[u])(\mathbf{S}(y)) \\ &= (\uparrow^0[[u](s)])(\uparrow^{0+1}[u])(\mathbf{S}(y)) \end{aligned}$$

2. $n = m + 1$: By case analysis on x :

+1. $x = \mathbf{Z}$:

$$\begin{aligned}
(\uparrow^n [u])(\uparrow^n [s])(x) &= (\uparrow^{m+1} [u])(\uparrow^{m+1} [s])(\mathbf{Z}) \\
&= (\uparrow^{m+1} [u])(\uparrow(\uparrow^m [s]))(\mathbf{Z}) \\
&= (\uparrow^{m+1} [u])(\mathbf{Z}) \\
&= (\uparrow^{m+1} [u])(\mathbf{Z}) \\
&= (\uparrow(\uparrow^m [u]))(\mathbf{Z}) \\
&= \mathbf{Z} \\
&= (\uparrow(\uparrow^m [[u](s)]))(\mathbf{Z}) \\
&= (\uparrow^{m+1} [[u](s)])(\mathbf{Z}) \\
&= (\uparrow^{m+1} [[u](s)])(\mathbf{Z}) \\
&= (\uparrow^{m+1} [[u](s)])(\uparrow(\uparrow^{m+1} [u]))(\mathbf{Z}) \\
&= (\uparrow^{m+1} [[u](s)])(\uparrow^{m+2} [u])(\mathbf{Z}) \\
&= (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(x)
\end{aligned}$$

+2. $x = \mathbf{S}(y)$:

$$\begin{aligned}
(\uparrow^n [u])(\uparrow^n [s])(x) &= (\uparrow^{m+1} [u])(\uparrow^{m+1} [s])(\mathbf{S}(y)) \\
&= (\uparrow^{m+1} [u])(\uparrow(\uparrow^m [s]))(\mathbf{S}(y)) \\
&= (\uparrow^{m+1} [u])(\mathbf{r}(\mathbf{S}, (\uparrow^m [s])(y))) \\
&= (\uparrow(\uparrow^m [u]))(\mathbf{r}(\mathbf{S}, (\uparrow^m [s])(y))) \\
&= \mathbf{r}(\mathbf{S}, (\uparrow^m [u])(\uparrow^m [s])(y))) && \text{by lemma 7} \\
&= \mathbf{r}(\mathbf{S}, (\uparrow^m [[u](s)])(\uparrow^{m+1} [u])(y))) && \text{by IH} \\
&= (\uparrow(\uparrow^m [[u](s)]))(\mathbf{r}(\mathbf{S}, (\uparrow^{m+1} [u])(y))) && \text{by lemma 7} \\
&= (\uparrow^{m+1} [[u](s)])(\mathbf{r}(\mathbf{S}, (\uparrow^{m+1} [u])(y))) \\
&= (\uparrow^{m+1} [[u](s)])(\uparrow(\uparrow^{m+1} [u]))(\mathbf{S}(y)) \\
&= (\uparrow^{m+1} [[u](s)])(\uparrow^{m+2} [u])(\mathbf{S}(y)) \\
&= (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(x)
\end{aligned}$$

□

Lemma 11 (Substitution lemma). $(\uparrow^n [u])(\uparrow^n [s])(t) = (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(t)$.

In particular, when $n = 0$, we have $[u]([s](t)) = [[u](s)](\uparrow [u])(t)$.

Proof. By induction on t :

1. $t = \underline{x}$:

$$\begin{aligned}
(\uparrow^n [u])(\uparrow^n [s])(\underline{x}) &= (\uparrow^n [u])(\uparrow^n [s])(x) \\
&= (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(x) && \text{by lemma 10} \\
&= (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(\underline{x})
\end{aligned}$$

2. $t = \lambda r$:

$$\begin{aligned}
(\uparrow^n [u])(\uparrow^n [s])(\lambda r) &= (\uparrow^n [u])(\uparrow^n [s])(\lambda r) \\
&= (\uparrow^n [u])(\lambda (\uparrow^{n+1} [s])(r)) \\
&= \lambda (\uparrow^{n+1} [u])(\uparrow^{n+1} [s])(r) \\
&= \lambda (\uparrow^{n+1} [[u](s)])(\uparrow^{n+2} [u])(r) && \text{by IH} \\
&= (\uparrow^n [[u](s)])(\lambda (\uparrow^{n+2} [u])(r)) \\
&= (\uparrow^n [[u](s)])(\uparrow^{n+1} [u])(\lambda r)
\end{aligned}$$

3. $t = @ (t_1, t_2)$: Straightforward by IH.

□